

For questions 1-3, factor each quadratic expression using the method of your choice.

1. $4b^2 + 8b - 45$

$(2b - 5)(2b + 9)$

2. $12x^2 + 7x - 49$

$(3x + 7)(4x - 7)$

3. $10d^2 - 9d + 2$

$(2d - 1)(5d - 2)$

For questions 4-5, consider the trinomial $9x^2 - 27x + 14$.

4. Write an expanded form of the trinomial that can be used to factor the expression by grouping.

$(9x^2 - 6x) + (-21x + 14)$

5. Factor the expression by grouping.

$$3x(3x - 2) - 7(3x - 2)$$

$$(3x - 7)(3x - 2)$$

For questions 6-7, consider the trinomial $6x^2 - 7x - 20$.

6. Complete the area model.

	$3x$	<u>$+4$</u>
$2x$	$6x^2$	<u>$8x$</u>
<u>-5</u>	<u>$-15x$</u>	-20

7. What are the two binomial factors for $6x^2 - 7x - 20$?

$(3x + 4) \text{ and } (2x - 5)$

For questions 8-9, consider the trinomial $8t^2 + 18t - 5$.

8. Complete the area model.

	<u>$4t$</u>	<u>-1</u>
<u>$2t$</u>	<u>$8t^2$</u>	<u>$-2t$</u>
<u>$+5$</u>	<u>$20t$</u>	<u>-5</u>

9. What are the two binomial factors for $8t^2 + 18t - 5$?

$(4t - 1)$ and $(2t + 5)$

10. Factor $10x^2 - 3x - 1$ using factoring by grouping.

$(10x^2 - 5x) + (2x - 1)$
 $5x(2x - 1) + 1(2x - 1)$
 $(5x + 1)(2x - 1)$